Hieu Man

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Summary

NLP researcher and Ph.D. candidate in Computer Science at the University of Oregon (expected 2027), two-time Amazon Applied Scientist intern, and author of 12 peer-reviewed papers (10 as first author) at venues including ACL, SIGIR, EMNLP, and AAAI. My research is on task-aware information modeling: enabling AI systems to identify the information a task depends on, represent it so it stays reliable when language, genre, or topic shifts, and reason over it across many steps. When any step fails, a system can answer confidently from the wrong information, a main reason AI is still hard to trust in high-stakes settings. I want to build retrieval and reasoning systems that learn which evidence to find, keep, and act on.

Education

University of Oregon — Eugene, OR
Ph.D. in Computer Science — *03/2023 - Present*
- Advisor: Prof. Thien Huu Nguyen. Expected graduation: Spring/Summer 2027.

University of Oregon — Eugene, OR
M.S. in Computer Science — *03/2023 - 10/2025*
- Thesis: *Improving Text Representations with Large Language Models* (ULLME, LUSIFER).

Hanoi University of Science and Technology — Hanoi, Vietnam
B.S. in Information Systems — *09/2016 - 04/2021*

Research & Industry Experience

Amazon AWS — New York, NY
Applied Scientist Intern — *06/2026 - 09/2026*
- **FORK: agent guardrails that explain rejections safely.** Built a guardrail for tool-using LLM agents whose rejection messages help the agent recover without helping the attacker. On FORK-World (677 adaptive-attack scenarios), it cut attack success from 78.4% to 14.1% on average (best baseline: 55.0%) while keeping utility above 85%; on MT-AgentRisk, with environments and tasks unseen in training, from 70.6% to 9.1%.
- **Gradient-free co-evolution.** Co-evolved the guardrail's decision and disclosure skills, kept as a natural-language library, with an attacker that reads every rejection message. Learned from 65 training pairs without weight updates, the frozen library transfers to unseen tasks, environments, and guardrail LLMs. Also built the FORK-World benchmark.

Amazon AWS — New York, NY
Applied Scientist Intern — *06/2025 - 09/2025*
- **State-Aware RAG.** Built a trainable working memory (Qwen3-4B extractor) that filters and updates retrieved evidence for multi-hop reasoning, with the retriever and generator frozen. Beat the best memory-augmented and RL-enhanced baselines by 8.6% and 9.3% on average across eight QA benchmarks, six unseen in training (Findings of ACL 2026; public code).
- **RL post-training with a Path-Outcome Dual Reward.** Designed a reward that also scores every reasoning step with an LLM judge, after an outcome-only reward credited lucky guesses. Trained the extractor with LoRA SFT on about 70K MCTS-synthesized state-action pairs, then with GRPO in verl, using LangGraph for multi-step rollouts on multi-GPU clusters.

University of Oregon — Eugene, OR
Research Assistant — *03/2023 - Present*
- **LLM embeddings for retrieval.** Built ULLME (EMNLP 2024 Demo; open-source, 1.5B to 8B checkpoints), which unifies contrastive and generation-augmented training and beats LLM2Vec trained on the same data by 1.2 to 2.0 points on MTEB. LUSIFER (SIGIR 2025) adapts an English-centric embedding model to other languages with English-only training, beating E5-Mistral by 3.19 points on average across 14 languages.
- **Agentic RAG over graph and text memory.** Built agentic RAG systems with an explicit working memory for long-horizon reasoning. Co-E (preprint, 2026) keeps graph and text memory in sync without any training; on an 8B open model, it beats comparable training-free baselines on six multi-hop QA benchmarks (+10.3 EM over HopRAG with GPT-4o on MuSiQue).
- **Robust representations under distribution shift (IARPA HIATUS).** Built the style embedding model that RTX BBN Technologies, the team lead, selected for the team's submission to the program's evaluations. Developed authorship attribution and machine-generated text detection methods robust across genres, topics, and languages: counterfactual augmentation (SIGIR 2024; released as ERLAS, 3,900+ downloads on Hugging Face) and EAVAE (ACL 2026; state of the art on Amazon Reviews, PAN21, and HRS).
- **Event relations and causality.** Developed document-level event relation and event causality identification methods that capture long-range context, using optimal transport (LREC-COLING 2024) and diffusion models (AAAI 2024).

VinAI Research — Hanoi, Vietnam
NLP Research Resident — *08/2021 - 03/2023*
- **Research shipped as a product.** Synthesized a grammatical error correction dataset, trained an error-detection model, and worked with a product team to ship it as a Microsoft Word add-in.
- **Document-level information extraction.** Built encoders for document-level event-event relation extraction and generation-based models for event causality identification.

Selected Publications & Preprints

- **Man Hieu** and Thien Huu Nguyen. "Co-Evolving Graph and Text Memory for Training-Free Multi-Hop Question Answering." *Preprint, arXiv:2607.23278* (2026).
- **Man Hieu**, Van-Cuong Pham, Nghia Trung Ngo, Franck Dernoncourt and Thien Huu Nguyen. "Explainable Disentangled Representation Learning for Generalizable Authorship Attribution in the Era of Generative AI." *Annual Meeting of the Association for Computational Linguistics* (2026).
- **Man Hieu**, Ro-ee Tal, Abhishek Kumar, Jaejin Cho and Benjamin Hsu. "Reasoning with Memory: Adaptive Information Management for Retrieval-Augmented Generation." *Findings of the Association for Computational Linguistics: ACL* (2026).
- **Man Hieu**, Nghia Trung Ngo, Viet Dac Lai, Ryan A. Rossi, Franck Dernoncourt and Thien Huu Nguyen. "LUSIFER: Language Universal Space Integration for Enhanced Multilingual Embeddings with Large Language Models." *International ACM SIGIR Conference on Research and Development in Information Retrieval* (2025).
- **Man Hieu**, Franck Dernoncourt and Thien Huu Nguyen. "Mastering Context-to-Label Representation Transformation for Event Causality Identification with Diffusion Models." *AAAI Conference on Artificial Intelligence* (2024).
- **Man Hieu**, Nghia Trung Ngo, Franck Dernoncourt and Thien Huu Nguyen. "ULLME: A Unified Framework for Large Language Model Embeddings with Generation-Augmented Learning." *Conference on Empirical Methods in Natural Language Processing: System Demonstrations* (2024).
- **Man Hieu**, Chien Van Nguyen, Nghia Trung Ngo, Linh Ngo, Franck Dernoncourt and Thien Huu Nguyen. "Hierarchical Selection of Important Context for Generative Event Causality Identification with Optimal Transports." *Joint International Conference on Computational Linguistics, Language Resources and Evaluation* (2024).
- **Man Hieu** and Thien Huu Nguyen. "Counterfactual Augmentation for Robust Authorship Representation Learning." *International ACM SIGIR Conference on Research and Development in Information Retrieval* (2024).
- Nguyen Thuat, Chien Van Nguyen, Viet Dac Lai, **Hieu Man**, Nghia Trung Ngo, Franck Dernoncourt, Ryan A. Rossi and Thien Huu Nguyen. "CulturaX: A Cleaned, Enormous, and Multilingual Dataset for Large Language Models in 167 Languages." *Joint International Conference on Computational Linguistics, Language Resources and Evaluation* (2024).
- **Man Hieu**, Nghia Trung Ngo, Linh Ngo Van and Thien Huu Nguyen. "Selecting Optimal Context Sentences for Event-Event Relation Extraction." *AAAI Conference on Artificial Intelligence* (2022).
- **Man Hieu**, Minh Nguyen and Thien Nguyen. "Event Causality Identification via Generation of Important Context Words." *Joint Conference on Lexical and Computational Semantics* (2022).
- **Man Hieu**, Duc Trong Le, Amir Pouran Ben Veyseh, Thuat Nguyen and Thien Huu Nguyen. "Introducing a New Dataset for Event Detection in Cybersecurity Texts." *Conference on Empirical Methods in Natural Language Processing* (2020).

Technologies

Languages & Frameworks: C/C++, Python, PyTorch, JAX/Flax, verl
Libraries & Tools: Git, Bash, SLURM, Hugging Face, vLLM, SGLang

Honors & Awards

Erwin & Gertrude Juilfs Scholarship in Computer and Information Science - University of Oregon 2026

Awarded to students who show exceptional promise for achievement, as evidenced by originality of research.

Gurdeep Pall Graduate Student Fellowship - University of Oregon 2024

Awarded for the overall quality of academic work, commitment to learning, and potential for further academic achievement.

First Prize - Relation Extraction Task, VLSP 2020 (Vietnamese Language and Speech Processing) 2020

Rank 1 in the first Vietnamese relation extraction shared task: 66.16% macro-F1, 2.74 points ahead of the runner-up.